

ARDUINO BASED REAL-TIME HOME CONTROL SYSTEM: DESIGN AND IMPLEMENTATION

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ABSTRACT

This paper presents the design and prototype implementation of an Arduino based real-time home control system that can be used to control and monitor home appliances remotely via the Internet. In this system, remote controlling and monitoring of four devices: TV, clothes washer, lamp, and Fridge are achieved by using two applications, mobile application and web application, which are developed from scratch. Results show that our proposed system allows a variety of devices to be controlled in a simple and interactive way, provides power consumption monitoring of devices to ensure the controlling process, and stores real-time data to the cloud in order to be accessed anywhere and anytime.

KEY WORDS: Home control system, Arduino platform, Web application, Mobile application, Cloud computing, Internet of Things.

1. INTRODUCTION

Technology covers many aspects of human everyday activities and it is everywhere. One form of using technology is to make our life easier and smarter, this is obvious in developing automation systems that help to monitor and control different types of devices that are used in industry, farms, military, and homes (Kumar and Abdul Qadeer. 2012). In home automation systems, many devices and software applications are integrated together to achieve the automation process (Javale et al. 2013). These systems are developed in different ways using different means to achieve two main goals: “controlling” and “monitoring” home appliances remotely. These systems are very useful especially to help handicapped and aged people to use their home appliances in an easy and smarter way (Sagar and Kusuma 2015). As we are in the era of connected devices, Internet of Things (IoT) is the main enabler of controlling and monitoring devices remotely. The IoT can be defined as the process of connecting everything around us to the Internet and manage their interactions (Khan et al. 2012). The core of any home automation system is a microcontroller with compatible sensors and actuators. Usually, the controlling process is done using actuators such as relays and the monitoring

process is performed using sensors such as temperature sensors (Banerjee and Singhal 2010). In order to interact with the hardware components of any home automation system, a medium of communication is required. Wireless communication such as GSM, ZigBee, Bluetooth, IR, and WiFi are widely used nowadays in this respect. Using the appropriate wireless technology is subject to many factors such as flexibility, efficiency, cost, accuracy, speed, and security (Ikram and Thornhill 2010). The proposed system extends the functionality of many previous works and presents its own set of unique features as presented in the subsequent sections.

As in Section 1 an introduction of this paper has been given, the remainder is organized as follows. Section 2 presents related works in literature and the proposed system is compared to other relevant works to highlight its features. The proposed system design and its prototype implementation are presented in section 3. In Section 4, the experimental results of the proposed system are shown and discussed. Finally, some final conclusions and areas for future work are provided in Section 5.

2. RELATED WORKS

The research community presented many approaches of designing and implementing home control systems. Some approaches combine home control with home security as well. This kind of combination is outside the scope of this paper. However, in this section we present briefly the most relevant works to our proposed approach with specifying their notable drawbacks. Authors (Yusuf and Aminu 2014) proposed a home automation system based on Arduino, ZigBee, and Android mobile application to control and monitor home devices through home gateway. This system has used many extra devices to achieve its goals and this makes its cost unfeasible. Also, it does not support accessing the system using non Android mobile application. Author (Mahesh 2014) presented a GSM based control system with Android mobile application. This system uses SMS messages based mobile application to control home devices. Major issues of this system are: no monitoring is considered, adds extra cost by using SMS messages as controlling commands, and it does not provide a user-friendly interface to interact the system in easy and interactive way. Authors (Bader et al. 2014) presented a design and implementation of a wireless real-time home automation system based on Arduino platform. Their proposed system offers two working modes: manually-automated mode for controlling using desktop application connected to a server and self-automated mode for controlling based on sensor values. This system does not allow users to interact with its functionality using their smart phones for instance, instead it has its own controlling complex interface which has been designed to help in this respect but this interface adds extra cost and complexity to the system. Authors (Ahmed et al. 2011) proposed a feature-rich Digital Home System (DHS). The DHS is controlled using voice commands and a mobile application. User voice commands are issued to a computer system which is in turn control home devices. Major issues of this system are voice commands are not often recognized correctly which cause many problems in operation and processing voice commands consume time. Authors (Subhi and Hajar 2014) proposed an Arduino based system to remotely control and

monitor electrical devices/sensors to provide power saving and home security. Standard mobile SMS messages with complex message format are used for the controlling process which adds extra cost to the system. This system does not provide a user-friendly interface to interact with the system. Authors (Kantharia et al. 2014) proposed the design of sensor based fault detection and remote monitoring system for measuring temperature. The aim of this system is to ensure that simple faults can be detected by sensors at an early stage and necessary actions could be taken so that they do not cause any harm or develop into serious failure. The main drawback of their system is that there is no controlling process over devices under monitoring.

After presenting many previous works which are related to the subject of our research, a comparative analysis has been conducted by the authors as shown in Table 1, to highlight the features of the proposed system compared to other relevant works in literature. This analysis shows many new features that have been added to this work which they were not presented by previous works. Such features, heterogeneous access, which means allowing users to interact with the system using different type of devices. For instance, users can use their Android mobile phones or non-Android mobile phones via web application. Notification messages, considered very important feature in case of facing an issue while using the system. These messages are triggered when there is no Internet connectivity, a specific device is not responding to user requests, and many others. Providing user-friendly interface is also taken into account while designing this system. This feature allows users to interact with the system in a simple and interactive way to get benefit from its functionality. As multiple users can interact with the system simultaneously, Application synchronization feature has been provided. This feature always displays the latest change has been done by any user. As a result, users always get the latest real-time states of all devices connected to the system. The system also provides cloud storage feature which stores the working states and power consumption of devices to be accessed by users anytime and anywhere.

Table (1): State of the art analysis

Factors	Articles	Sagar and Kusuma 2015	Yusuf and Aminu 2014	Maresh 2014	Bader el at. 2014	Subhi and Hajar 2014	Kantharia el at. 2014	Ramlee el at. 2013	Baig el at. 2013	Bhaskar and Swarnalatha 2015	Piyare and Ro 2013	Salunke el at. 2013	Yeole el at. 2015	The proposed work
Goal	Controlling	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓
	Monitoring	✓	✓		✓	✓	✓	✓		✓	✓	✓	✓	✓
Remoter	Computer	✓					✓	✓						✓
	Mobile	✓	✓	✓	✓	✓		✓	✓	✓	✓	✓	✓	✓
Communication	Internet	✓	✓		✓		✓	✓			✓	✓		✓
	GSM			✓		✓			✓					
	ZigBee		✓						✓					
	Bluetooth							✓	✓	✓			✓	
Mobile Application			✓	✓				✓	✓	✓	✓	✓		✓
Web Application		✓					✓							✓
Web Service											✓			✓
Cloud Storage		✓										✓		✓
Heterogeneous access		✓					✓							✓
Application Synchronization														✓
Accuracy/Reliability		✓	✓			✓	✓	✓		✓	✓	✓	✓	✓
Low Cost		✓						✓		✓	✓	✓	✓	✓
Easy installation and Configuration		✓		✓				✓		✓	✓	✓	✓	✓
Reusability		✓		✓					✓		✓	✓	✓	✓
Notification Messages							✓							✓
User-friendly Interface			✓	✓				✓		✓	✓	✓		✓

3. SYSTEM DESIGN AND IMPLEMENTATION

In order to understand the internal work of the proposed system, we list out the software requirements of the system. Then, both system modules and hardware components are explained in this section respectively.

3.1. SOFTWARE REQUIREMENTS

A number of programming and scripting languages have been used in the implementation of the system as follows:

- **Arduino programming language**
It has been used to program the core functionality of the Arduino boards and compatible sensors and actuators. It is worth mentioning that Arduino IDE (Integrated Development Environment) has been used in this respect.
- **Java programming language**
It has been used to program the mobile application. Android Studio IDE has been used in this respect.
- **JavaScript scripting language**
It has been used to program the functionality of the web application alongside HTML (Hyper Text Markup Language) and CSS (Cascading Style Sheet) to create and style the pages of the web application respectively.

3.2. SYSTEM'S MODULES

The proposed system consists of several modules, as shown in Figure 1. Organizing the system design into modules provides better understanding and easiest way of managing and handling unexpected problems during the system works. Furthermore, it makes the future software maintenance more easier if required.

- **Remoter Module**
Through this module, users can use a mobile phone or a computer to interact with the proposed system. Two applications have been developed from scratch to help users in this respect, an Android mobile application and a web application. It is worth mentioning that using mobile devices to interact with the system is more convenient compared to other available options because of their small size and their mobility. On top of that users can use the mobile application or the web

application on the same device to access the functionality of the system.

- **Communication Module**

The Internet is used as a communication medium that connects remoter module to the core of the system. This enables users to access home network directly when they are within the range of it or indirectly via public network when they are outside home.

- **Controlling Module**

Arduino platform and specifically Arduino Yun has been used to do the controlling and monitoring processes. Arduino Yun is used widely for connecting IoT devices to the Internet and developing IoT based applications. It is a good choice in this respect as it has built-in Ethernet, built-in WiFi, internal storage, and supports external storage via micro-SD. On top of that, it provides a number of digital pins which are 13 pins that support either 0 (0V) or 1(5V) input/output operations and analog pins which are 6 pins that support only input values between 0(0V) and 1023(5V). These different pins support reading/writing from/to compatible devices in order to manage their functionalities.

- **Sensors and Actuators Module**

A 4 channel 5V relay module has been used for switching home devices ON/OFF. This module works from a 5V supply for the relay and 0V or 5V for the switching signal of the 4 independently controlled channels. It uses a transistor and opto-isolator to switch the channels ON and OFF. It is worth mentioning that each channel switches up to 10Amps and up to 250V. Besides, ACS712 current sensor has been used for measuring power consumption of the used devices. This sensor works from a 5V supply and its measurement range is between -5 and +5 Amps. Its working details are given in subsection 3.3 alongside the required equations that have been used to convert the sensor measurements to Arduino-acceptable values.

- **Household Appliances Module**

Many home devices have been used in the experimental testing of the proposed system such as TV, lamp, washer, and fridge to ensure the proper working of the system. Users are able to control these devices in an easy and interactive way.

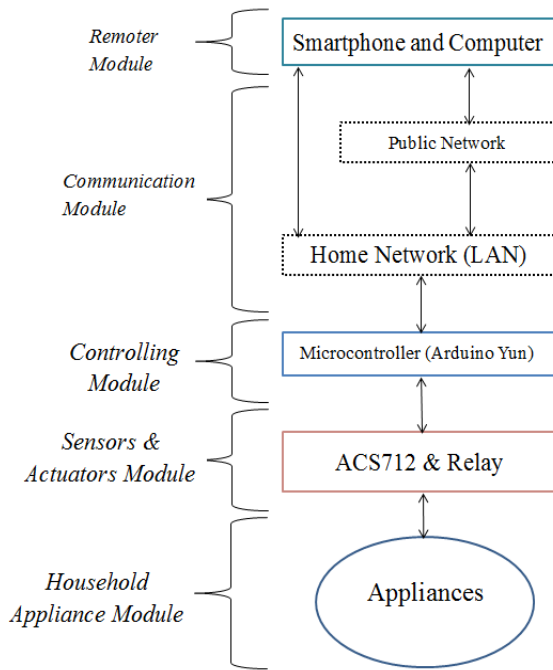


Fig. (1): System's modules

3.3. SYSTEM'S HARDWARE COMPONENTS

The hardware components of the proposed system and their connections are shown in Figure 2.

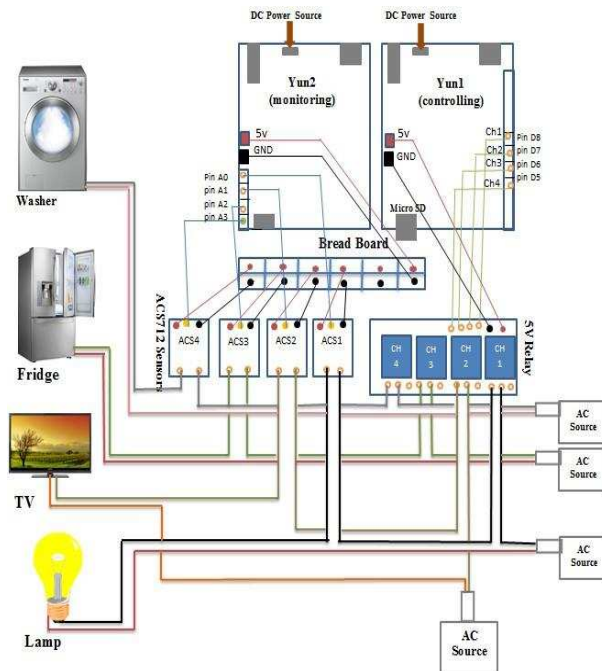


Fig. (2): System's hardware components

The proposed system can be installed easily in any home or can be considered while constructing new smart homes. In this system two Arduino Yun boards are used (Yun1 for controlling and

Yun2 for monitoring) to overcome the issue of hardware level threading as most microcontrollers do not support threading feature. Threading is required in our system to accomplish two tasks (controlling and monitoring) simultaneously instead of doing one task then waiting for a while to do the second task.

Yun1 controls household devices using a relay compatible with Arduino and has four channels to control four devices separately; each channel passes up to 10A of current. Table 2 shows Yun1 digital pins connected to relay's channels to control four devices.

Table 2: Yun1 pin connections

Digital Pins	Relay Channels	Household Devices
D8	CH1	Lamp
D7	CH2	TV
D6	CH3	Fridge
D5	CH4	Washer

Yun2 measures and monitors the power consumption of household devices using four 5A current sensors. The measured power consumption values are then stored in the cloud (Google spreadsheets) using Temboo APIs. Table 3 shows Yun2 analog pins connected to four current sensors to monitor four devices.

Table 3: Yun2 pin connections

Analog Pins	Current Sensors	Household Devices
A0	ACS1	Lamp
A1	ACS2	TV
A2	ACS3	Fridge
A3	ACS4	Washer

Measuring power consumption of each device ensures the controlling process. For example, when users switch a device OFF, its power consumption goes down. So, it has been decided to make connections between relay's channels and the corresponding current sensors as shown in Table 4.

Table 4: Relay's channels and current sensors connections

Relay channel	ACS712 sensor
CH1	ACS1
CH2	ACS2
CH3	ACS3
CH4	ACS4

The current sensor used in this system detects current as values between 0 (0V) and 1023 (5V) that is 0.0049V for each increment via one of Yun2 analog input pins. As Arduino can only handle voltage values between 0V and 5V, current sensor values can be converted into an acceptable voltage by using Equation 1.

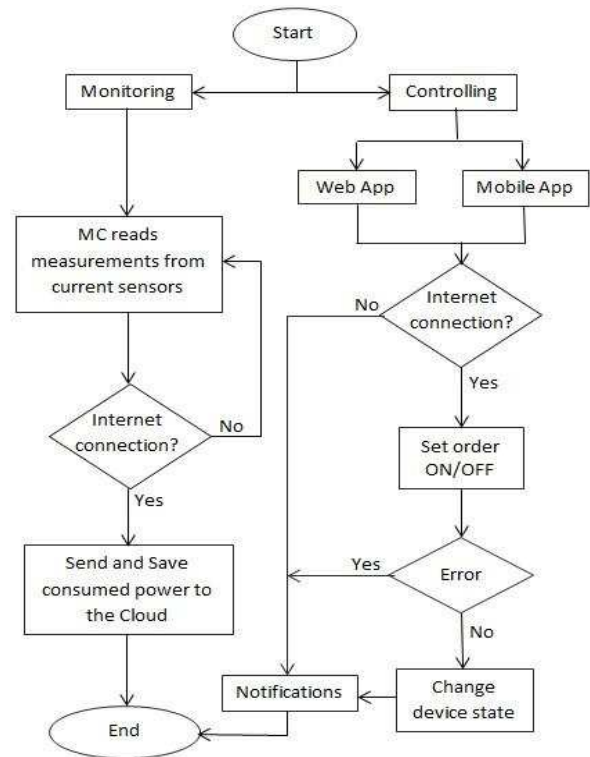
$$\text{Voltage (V)} = \text{raw value} / 1023 * 5 \dots\dots\dots (1)$$

The sensitivity of the used 5A sensor is 0.185V/A and the sensor can measure positive and negative currents range between -5A and 5A, so if input current is 0, output value is 2.5V. To convert the current sensor raw values to Current, Equation 2 has been used.

$$\text{Current (I)} = (0.0049 * \text{raw value} - 2.5) / 0.185 \dots (2)$$

Then, power consumption (in Watts) is calculated for each device from the obtained current (in Amps). The calculated values are then stored in the cloud (Google spreadsheets) to enable users to ensure the controlling process and to monitor their home devices anytime and anywhere.

Before discussing the system scenarios and its testing, we provide Figure 3 below that presents the operational flow of the system and how can be used.

**Fig. (3):** System's flowchart

4. EXPERIMENTAL RESULTS AND DISCUSSION

The results of this system are given in this section as scenarios. Scenarios are good choice to illustrate the practical part of the system and make the system's functionality more understandable.

4.1. Scenario 1: Interaction with web app.

The web application that has been developed as part of the system is shown in Figure 4; it provides a simple graphical user interface to enable users switch their home devices ON/OFF and monitor them remotely via the Internet. Users will only interact with the web application after they register their information then login to the application. This login process secures the access to the web application.

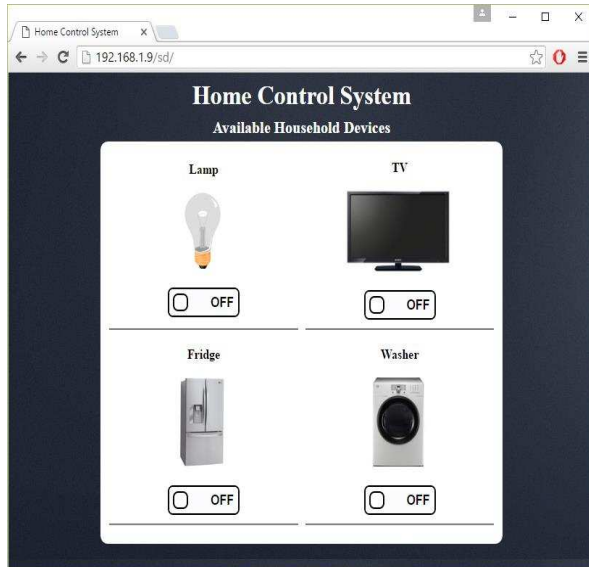


Fig. (4): Web application user interface

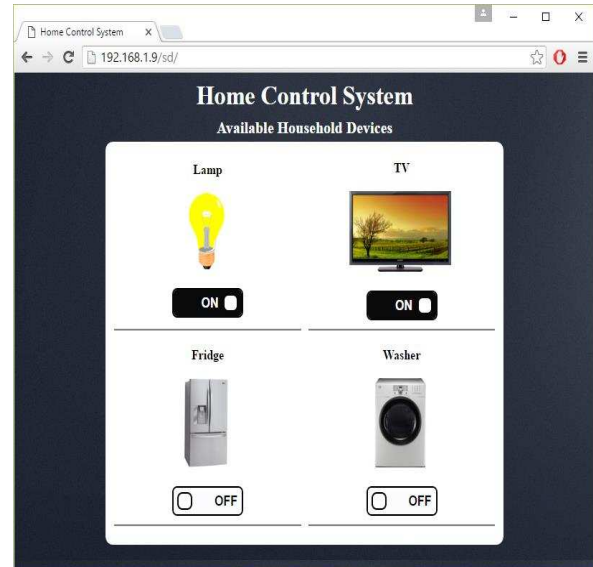


Fig. (5): Web application with ON/OFF states

The web application shows ON/OFF states of four devices used in the system. Each device is identified by its name, picture, and ON/OFF button to control its state. To control the state of any device, users simply click on the button related to the device. Figure 5, shows the web application with some devices in the ON state. When the state of any device is changed by clicking on its button, three things are changed accordingly: the button text, background color, and the device picture to colorful one. Changing these things help users to ensure that the controlling process is working properly. In case of no change in them, that means the controlling process is not performed as requested by users for some reasons such as no Internet connectivity, the system is not turned on, or the device itself is not working properly. However, users will be notified in such cases by notification messages displayed for them.

4.2. Scenario 2: Interaction with mobile app.

Alongside the web application, a mobile application has been developed to add more flexibility. This mobile application is very easy to be used with a simple user interface which allows users to control and monitor their home devices as shown in Figure 6.

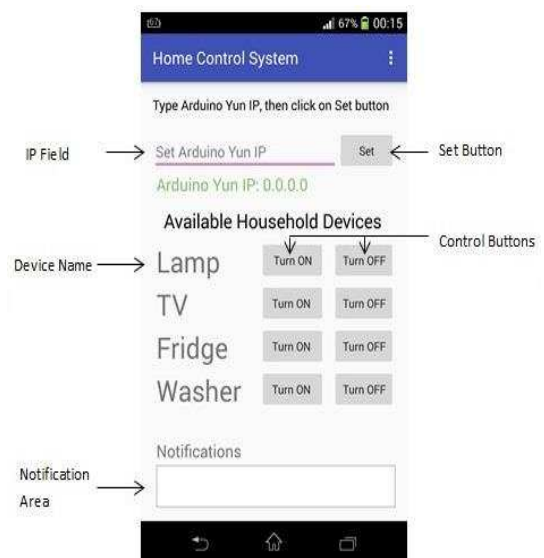


Fig. (6): Mobile application user interface

After using the static IP address of the system with the availability of Internet, users should be able to control their home devices remotely. In case of any problem, users will be notified on the notification area in the mobile application. Figure 7 shows some cases of actual interaction with the system.

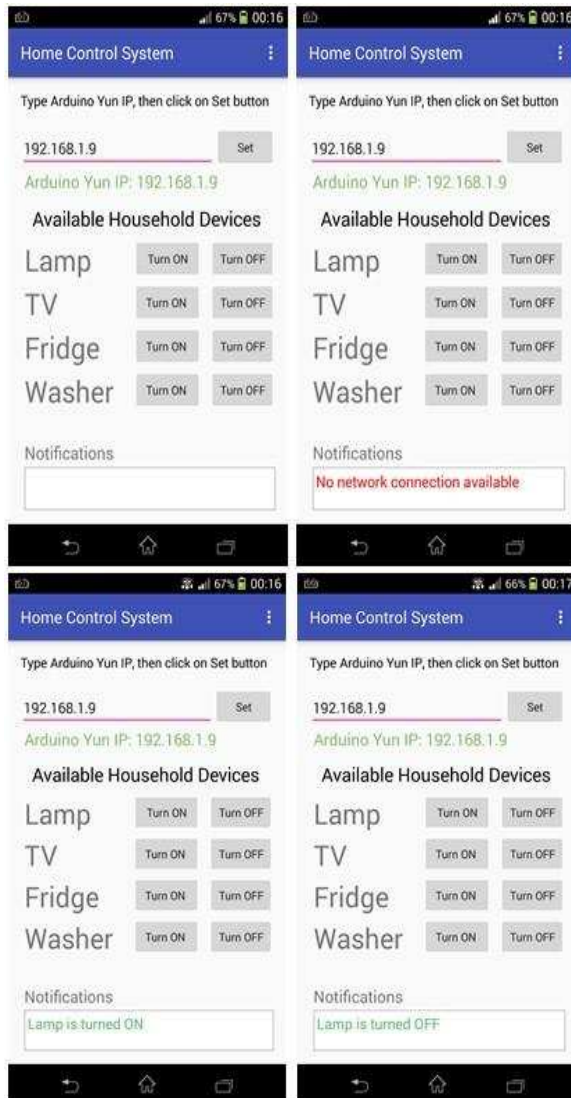


Fig. (7): Mobile application with different working states

Both the web application and mobile application use REST (REpresentational State Transfer) URL to control home devices remotely using Internet. REST URL used format is shown below:

`http://<device-address>/<type>/<state>/<pin>/<writing>`

device-address: represents static IP or name of the used microcontroller.

type: represents the type of hardware platform.

state: represents function calling state.

pin: represents the used pin.

writing: represents digital/analog writing value.

The example below shows switching a lamp ON and OFF respectively:

<http://ArduinoIP/arduino/digital/8/1>

<http://ArduinoIP/arduino/digital/8/0>

The rationale behind developing the web application alongside Android mobile application is sometimes users may use non-Android devices to interact with the system such as iPhone mobiles and iPad tablets which are using iOS operating system. In this case, users will not be able to use our Android mobile application because it can be used only with any device uses Android operating system. The web application helps in this respect as a generic application which can be accessed and used by heterogeneous devices using different operating systems. Another important real-time feature of the proposed system is the synchronization between the web application and mobile application. For example, any controlling process using mobile application updates the display in the web application, this process called application synchronization. So, when users turn on a lamp, the lamp state in the web application will be changed too. This synchronization is very important for users to always get the latest states of their devices as sometimes more than one user may try to interact with the system simultaneously.

5. CONCLUSION AND FUTURE WORK

In this paper, an Arduino based real-time home control system is proposed and implemented to enable users control and monitor their home devices remotely using Internet. Comparing our system to previous works shows many advantages that our system provides. For future work, other important features can be implemented and added to the system, such as system's scalability, which allows users to add other new home appliances to the system without facing difficulties or changing the software design. Users should be able to easily add/plugin new devices then register them using new software feature "add new appliance" that should be provided by the web application and mobile application used in this system.

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